Linear Applications and Matrices

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Matrices and Linear Applications

Rank of a set of vectors

Definition

Let E be a Vector Space and $\{v_1, \dots, v_p\}$ a finite set of vectors of E. Then, the **rank** of the set $\{v_1, \dots, v_p\}$ is the dimension of the Vector Subspace $\mathrm{Vect}(v_1, \dots, v_p)$ generated by the vectors $v_1, \dots, v_p$. I.e.,

$$\boxed{\mathrm{rank}(v_1, \dots, v_p) = \dim\big(\mathrm{Vect}(v_1, \dots, v_p)\big)}$$

Proposition

Let E be a $\mathbb{K}$-Vector Space of finite dimension n and $\{v_1, \dots, v_p\}$ a set of p vectors of E. Then,

1. $0 \leq \mathrm{rank}(v_1, \dots, v_p) \leq p$,
2. $\mathrm{rank}(v_1, \dots, v_p) \leq n$.

Remarks :

- The rank of a set is 0 if and only if all the vectors in the set are zero-vectors.
- The rank of a set $\{v_1, \dots, v_p\}$ is p if and only the set is linearly independent.

Example :

Determine the rank of the set $\{v_1, v_2, v_3\}$ in the Vector Space $\mathbb{R}^4$, with

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad v_3 = \begin{pmatrix} (-1) \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

Dimension of the Vector Space $M_{np}(\mathbb{R})$

Proposition

The dimension of the set of matrices of size $n \times p$ is

$$\boxed{\dim M_{np}(\mathbb{R}) = np}$$

Rank of a matrix

Definition

We define the rank of a matrix as the rank of the set of its column vectors.

Example :

Determine the rank of the matrix

$$A = \begin{pmatrix} 1 & 2 & (-1/2) & 0 \\ 2 & 4 & (-1) & 0 \end{pmatrix}$$

Proposition

The rank of a column echelon form matrix (the number of zeroes at the beginning of successive columns is increasing) is the number of non-zero columns.

Proposition

The rank of a matrix whose columns are $C_1, C_2, \ldots, C_n$ is unchanged by the following operations :

- $C_i \leftarrow \lambda C_i$ (with $\lambda \neq 0$),
- $C_i \leftarrow C_i + \lambda C_j$ ($\lambda \neq 0$, $i \neq j$), and more generally $C_i \leftarrow C_i + \sum_{j \neq i} \lambda_j C_j$,
- $C_i \leftrightarrow C_j$.

How to determine the rank of a matrix or a set of vectors ?

Methodology

- Apply Gauss method on the columns of the matrix (whose columns are the vectors),
- Transform the matrix to its column echelon form,
- The rank is then the number of non-zero columns.

Example :

1) Determine the rank of the set $\{v_1, v_2, v_3, v_4, v_5\}$ in the Vector Space $\mathbb{R}^4$, with

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} (-1) \\ 2 \\ 0 \\ 1 \end{pmatrix} \quad v_3 = \begin{pmatrix} 3 \\ 2 \\ (-1) \\ (-3) \end{pmatrix} \quad v_4 = \begin{pmatrix} 3 \\ 5 \\ 0 \\ (-1) \end{pmatrix} \quad v_5 = \begin{pmatrix} 3 \\ 8 \\ 1 \\ 1 \end{pmatrix}$$

2) Consider the set $S = \{v_1, v_2, v_3, v_4\}$ in the Vector Space $\mathbb{R}^3$, with

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad v_2 = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} \quad v_3 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \quad v_4 = \begin{pmatrix} (-1) \\ 2 \\ 2 \end{pmatrix}$$

Show that the set S generates $\mathbb{R}^3$.

Theorem

A square matrix of size $n \times n$ is invertible if and only if it is of rank n.

Proposition

Let $A \in M_{np}(\mathbb{K})$ whose columns $v_1, \ldots, v_n$ and rows $w_1, \ldots, w_n$, then

- rank(A)=dim(Vect$(w_1, \ldots, w_n)$),
- the vector spaces generated by the vector columns Vect$(v_1, \ldots, v_n)$ and rows Vect$(w_1, \ldots, w_n)$ of A are of same dimension,
- rank(A) = rank(A^T).

Matrix of a linear application

Definition

- Let E and F be two Vector Spaces of finite dimensions n and p of bases $\mathcal{B} = \{e_1, \dots, e_p\}$ and $\mathcal{B}' = \{f_1, \dots, f_n\}$. Let $f : E \to F$ be a linear application. Then

$$\forall j \in \{1, \dots, p\}, \exists!(a_{1j}, \dots, a_{nj}) \in \mathbb{K}^n, f(e_j) = a_{1j} f_1 + \dots + a_{nj} f_n$$

- We thus define the **matrix of the linear application** f with respect to the bases $\mathcal{B}$ and $\mathcal{B}'$, of size $n \times p$ and denoted as $\mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f)$, as

$$\mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f) = \begin{pmatrix} a_{11} & \dots & a_{1p} \\ a_{21} & \dots & a_{2p} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{np} \end{pmatrix}$$

- $\mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f)$ is the matrix whose columns are the images by f of the vectors of $\mathcal{B}$ expressed in $\mathcal{B}'$.

Example :

Determine the matrix of the linear application $f : \mathbb{R}^3 \to \mathbb{R}^2$ defined as

$$f(x,y,z) = (x+y-z, x-2y+3z)$$

a) in the canonical bases $\mathcal{B}, \mathcal{B}'$,

b) in the bases $\mathcal{B}_0 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ and $\mathcal{B}_0' = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$.

Operations on linear applications and matrices

Proposition

Let $f, g : E \to F$ be two linear applications, $\mathcal{B}$ a basis of E and $\mathcal{B}'$ a basis of F. Then :

- $\text{Mat}_{\mathcal{B},\mathcal{B}'}(f+g) = \text{Mat}_{\mathcal{B},\mathcal{B}'}(f) + \text{Mat}_{\mathcal{B},\mathcal{B}'}(g)$,
- $\text{Mat}_{\mathcal{B},\mathcal{B}'}(\lambda \cdot f) = \lambda \cdot \text{Mat}_{\mathcal{B},\mathcal{B}'}(f)$.

Proposition

Let $f : E \to F$ and $g : F \to G$ be two linear applications, $\mathcal{B}$ a basis of E, $\mathcal{B}'$ a basis of F and $\mathcal{B}''$ a basis of G. Then :

$$\boxed{\text{Mat}_{\mathcal{B},\mathcal{B}''}(g \circ f) = \text{Mat}_{\mathcal{B}',\mathcal{B}''}(g) \times \text{Mat}_{\mathcal{B},\mathcal{B}'}(f)}$$

Exercise :

Let the two linear applications $f : E \to F$ and $g : F \to G$, with $E = \mathbb{R}^2$, $F = \mathbb{R}^3$ and $G = \mathbb{R}^2$.

Let $\mathcal{B} = (e_1, e_2)$ a basis of E, $\mathcal{B}' = (f_1, f_2, f_3)$ a basis of F and $\mathcal{B}'' = (g_1, g_2)$ a basis of G.

We consider furthermore that

$$A = \mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \text{ and } B = \mathrm{Mat}_{\mathcal{B}',\mathcal{B}''}(g) = \begin{pmatrix} 2 & (-1) & 0 \\ 3 & 1 & 2 \end{pmatrix}$$

Compute the matrix $C = \mathrm{Mat}_{\mathcal{B},\mathcal{B}''}(g \circ f)$ by means of two different methods.

Matrix of an endomorphism

Definition

The matrix of an endomorphism $f : E \to E$ is a matrix of size $n \times n$ denoted

- $\mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f)$: if we consider two different starting and ending bases,
- $\mathrm{Mat}_{\mathcal{B}}(f)$: if we consider the same starting and ending basis.

Examples :

- The endomorphism identity is of matrix $\mathrm{Mat}_{\mathcal{B}}(Id_E) = I_n$ if we consider the same starting and ending basis,
- The endomorphism homothecy $(h_\lambda(x) = \lambda \cdot x)$ is of matrix $\mathrm{Mat}_{\mathcal{B}}(h_\lambda) = \lambda I_n$ if we consider the same starting and ending basis,
- The endomorphism rotation of angle θ centered at the origin $(r_\theta(x, y) = (x \cos\theta - y \sin\theta, x \sin\theta - y \cos\theta))$ is of matrix
$$\mathrm{Mat}_{\mathcal{B}}(r_\theta) = \begin{pmatrix} \cos(\theta) & -sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

Corollary

Let E be a Vector Space of finite dimension, $\mathcal{B}$ a basis of E and $f : E \to E$ an endomorphism. Then

$$\forall p \in \mathbb{N},\ \mathrm{Mat}_{\mathcal{B}}(f^p) = \mathrm{Mat}_{\mathcal{B}}(\underbrace{f \circ \ldots \circ f}_{p \text{ times}}) = \left(\mathrm{Mat}_{\mathcal{B}}(f)\right)^p$$

Example :

Let r_θ be the rotation of angle θ in $\mathbb{R}^2$ centered at the origin. Determine the matrix of r_θ^p.

Matrix of an isomorphism

Theorem : Characterisation of the matrix of an isomorphism

Let E, F be two $\mathbb{K}$-Vector Spaces of same finite dimension and $f : E \to F$ a linear application. Let $\mathcal{B}$ a basis of E, $\mathcal{B}'$ a basis of F and $A =$Mat$_{\mathcal{B},\mathcal{B}'}(f)$. Then

1. f is bijective (then an isomorphism) if and only if A is invertible,
2. If f is bijective, then the matrix of $f^{-1} : F \to E$ is A^{-1}. I.e.,

$$\text{Mat}_{\mathcal{B}',\mathcal{B}}(f^{-1}) = \left(\text{Mat}_{\mathcal{B},\mathcal{B}'}(f)\right)^{-1}$$

Example :

Let $r : \mathbb{R}^2 \to \mathbb{R}^2$ the rotation of angle $\pi/6$ centered at the origin and s the reflection with respect to the line $y = x$. Determine the matrix of $(s \circ r)^{-1}$ in the canonical basis.

Change of basis

Link between linear application and matrices

Definition

- Let E be a Vector Space of finite dimension p and $\mathcal{B} = \{e_1, \ldots, e_p\}$ a basis of E. Then $\forall x \in E$, $x = x_1 e_1 + \ldots + x_p e_p$ and
$$\text{X}=\text{Mat}_{\mathcal{B}}(x) = \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix}_{\mathcal{B}}$$
- Let F be a Vector Space of finite dimension n and $\mathcal{B}' = \{f_1, \ldots, f_n\}$ a basis of F. Then $\forall y \in F$, $y = y_1 f_1 + \ldots + y_p f_p$ and
$$\text{Y}=\text{Mat}_{\mathcal{B}'}(y) = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix}_{\mathcal{B}'}$$
- Let $f : E \to F$ be a linear application and A=$\text{Mat}_{\mathcal{B},\mathcal{B}'}(f)$. If $y = f(x)$, then
$$\boxed{Y = A \times X} \quad \text{and} \quad \boxed{\text{Mat}_{\mathcal{B}'}(y) = \text{Mat}_{\mathcal{B},\mathcal{B}'}(f) \times \text{Mat}_{\mathcal{B}}(x)}$$

Exercise :

Let E be a Vector Space of dimension 3 with a basis $\mathcal{B} = (e_1, e_2, e_3)$. Let f be an endomorphism of E of matrix

$$\text{A=Mat}_{\mathcal{B}}(f) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Determine the kernel and the image of f.

2) Transformation (or transition) matrix

Definition

Let E be a Vector Space of finite dimension n and $\mathcal{B}$, $\mathcal{B}'$ two different bases of E. Then we call the **transformation (or transition) matrix** from the basis $\mathcal{B}$ to the basis $\mathcal{B}'$, denoted $T_{\mathcal{B},\mathcal{B}'}$, the square matrix of size $n \times n$ whose columns are composed of the coordinates of the vectors of $\mathcal{B}'$ expressed in the basis $\mathcal{B}$.

Example :

Determine the transformation matrix in $\mathbb{R}^2$ from the basis $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ to the basis $\mathcal{B}' = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 5 \\ 4 \end{pmatrix} \right\}$.

Proposition

Let E be a Vector Space of finite dimension n and of bases $\mathcal{B}$, $\mathcal{B}'$, $\mathcal{B}''$. Then

1. $\boxed{T_{\mathcal{B},\mathcal{B}'} = \mathrm{Mat}_{\mathcal{B}',\mathcal{B}}(Id_E)}$
2. $T_{\mathcal{B},\mathcal{B}'}$ is invertible and its inverse matrix is $T_{\mathcal{B}',\mathcal{B}}$: $\boxed{T_{\mathcal{B}',\mathcal{B}} = \left(T_{\mathcal{B},\mathcal{B}'}\right)^{-1}}$,
3. $\boxed{T_{\mathcal{B},\mathcal{B}''} = T_{\mathcal{B},\mathcal{B}'} \times T_{\mathcal{B}',\mathcal{B}''}}$
4. Let $x \in E$ with $X = \mathrm{Mat}_{\mathcal{B}}(x) = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}_{\mathcal{B}}$ and $X' = \mathrm{Mat}_{\mathcal{B}'}(x) = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}_{\mathcal{B}'}$,

   then
   $$\boxed{X = T_{\mathcal{B},\mathcal{B}'} \times X'}$$

Formula for basis change

Theorem

Let $f \in \mathcal{L}(E,F)$, $\mathcal{B}_E = \{e_1,\dots,e_n\}$, $\mathcal{B}'_E = \{e'_1,\dots,e'_n\}$ two bases of E and $\mathcal{B}_F = \{f_1,\dots,f_n\}$, $\mathcal{B}'_F = \{f'_1,\dots,f'_n\}$ two bases of F. Then if we denote

$$P = T_{\mathcal{B}_E,\mathcal{B}'_E},\ Q = T_{\mathcal{B}_F,\mathcal{B}'_F},\ A = \mathrm{Mat}_{\mathcal{B}_E,\mathcal{B}_F}(f) \text{ and } B = \mathrm{Mat}_{\mathcal{B}'_E,\mathcal{B}'_F}(f)$$

we obtain

$$\boxed{B = Q^{-1}AP}$$

Proposition

Let $f \in \mathcal{L}(E,E)$, $\mathcal{B}_E = \{e_1,\dots,e_n\}$, $\mathcal{B}'_E = \{e'_1,\dots,e'_n\}$ two bases of E. Then if we denote

$$P = T_{\mathcal{B}_E,\mathcal{B}'_E},\ A = \mathrm{Mat}_{\mathcal{B}_E}(f) \text{ and } B = \mathrm{Mat}_{\mathcal{B}'_E}(f)$$

we obtain

$$\boxed{B = P^{-1}AP}$$

Exercise :

Let the two bases

$$\mathcal{B}_1 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ (-1) \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ (-1) \end{pmatrix} \right\} \quad \text{and} \quad \mathcal{B}_2 = \left\{ \begin{pmatrix} 1 \\ (-1) \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ (-1) \end{pmatrix} \right\}$$

of $\mathbb{R}^3$.

Let f be the linear application of matrix in the basis $\mathcal{B}_1$

$$\text{A=Mat}_{\mathcal{B}_1}(f) = \begin{pmatrix} 1 & 0 & (-6) \\ (-2) & 2 & (-7) \\ 0 & 0 & 3 \end{pmatrix}$$

Determine the matrix of f in the basis $\mathcal{B}_2$.

Matrix similarity

Definition

Let A, B two matrices of $M_n(\mathbb{K})$. We say that the matrix B is **similar** to the matrix A if there exists an invertible matrix $P \in M_n(\mathbb{K})$ such that $B = P^{-1}AP$.

Proposition

Two similar matrices represent the same endomorphism but in two different bases.

Example :

Give two similar matrices for the linear application $f : (x, y) \mapsto (x - y, x + y)$.

END

References :

-*Joseph Griphone, ALGÉBRE LINÉAIRE*, 5e edition.
-*Arnaud Bodin et al., Exo7, Université de Lille*.